

ActInf GuestStream 050.1 ~ Valeria Becattini & Anna Ciaunica, "Selfless Minds, Unlimited Bodies?"

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hello and welcome it's July 27 2023 and we're here in actin guest stream number 50.1 with Valeria becatini and Anna C Unica this is going to be a very interesting and cool presentation and discussion on selfless Minds unlimited bodies homeostatic bodily self-regulation and meditative experiences there will be a presentation followed by a discussion so thank you both for joining and to you Valeria for the presentation thank you Diana very much I'm very happy to be here with Anna presenting a path of our [Music] um wider project is a paper about what happened in Southwest state from a homo suffering body self-regulation perspective so it's as I said it's part of a wider project that consists in both theoretical and experimental components and So today we're gonna look at meditation so for people that are not really familiar with that organization is attracted engaged attention to self-regulate physiological and cognitive activity and what we're interested here is that people that meditate with a certain experience so for example meditating in the past few years with regularly like some hours per week they can consistently report losing oneself uh one sense of the self or having selfless experiences but only not only that they also report to have uh bodily uh alterations so alterations in the procession of the body that has been called expansion Oceanic boundlessness dissolution of body boundaries and since you are particularly interested in this kind of phenomenal experience we have proposed to frame them as unlimited body just uh for clarity so our question here is what exactly and the question there was we'll follow the whole uh presentation in the background is what exactly exactly is lost when I experience many faces report losing their self and the body so let's first looks uh very briefly uh what what um the self is or the sounds of selfies so this content is so complex and so wide that as we debated since the beginning of philosophy um and of course it's so it might be so intrusive that it's also very hard to Define but one very old uh old and used definition is the unitary or unique individual I the first person perspective that laughing was called the subjectum so something that lay under or uh stream of experience thoughts and whatever else happens to us that's the

subject um however more recently in cognitive signs we are appealing to a more common definition of sense of self a bit more exhaustive that is um defined as a multi-dimensional construct and here uh we will come back on this on this concept later and we'll unpack it a bit more so the aim of this presentation will be to first of all lay down the fact that in self-experiences the biological organism so to speak the the human body remains successful that self-regulated and this uh might be very intuitive but it needs them to see that during selfless uh experiences what we propose is that they may be uh tacitly a compiler yeah implicitly underlined by enhanced connectedness with this successful self-regulation that we proposed to Lincoln with a perceptually attenuated body or what has been already called transparent body uh and this uh allows it to propose it's of course in a working process and we're very open to uh feedback at any stage of this presentation that the term selfless States may be misleading because what is lost is an ordinary way to mentally models the self relation to the body and the world so I wrote my is composed as uh as follows first I will um passing through uh debates and definitions of losing oneself and body so that means they'll go through alterations in these um three-dimension uh during that deep massive state and then I will introduce King concept of predictive process interactive info framework that are useful in our explanation so that then I can uh um unpack the role of the biological self-organization and homostatic self-regulation in constituting our sense of self and then Pastor or uh claim that the so much process is like some other sensory attenuation and transparent self modeling our key mechanism underlying the phenomenology of the Methodist States our conclusion and present uh ongoing related projects so let's back come back to the sensor itself so uh in the in when one has to uh investigate the sense of self of course it as I said it's a wide concept there so many definitions and um characterization of the sensor cell has come up uh in philosophy and cognitive science so we have uh terms like minimal self self experience self-hood sense of ownership and all of this and for author I think we have just to stick to a common definition that comes for example from from a similar paper from Gallagher to thousand that defines it as a multi-dimensional construct composed by minimal or embodied components and narrative or reflective components so to unpack a little bit these two concepts the minimal embodied experience of the sense of self is the basic um implicit functioning of the um by the YouTube of the ongoing moment so the stream of experience that happens uh to me right now uh can be defined as my minimal sense of self so I don't really need to have an explicit knowledge about myself to have this kind of Dimension uh and these uh can be further unpacked through the accounts of our pre-reflective self-consciousness that comes uh from a phonological tradition tracing back

from uh about to uh uh argue that uh parts of our self-conscious does not does not really need any reflection to to happen there or in place an explicit thought it happens um in a way tambrian uh is conservatively before uh reflection and then on top of this we build our narrative or our Reflections about ourselves and this is more linked to uh self-identification processes or self-monitoring self-recognition and introspection introspection it's it's um the ability of introspect so perceive yourself or reflect about yourself in an explicit way so you do need to have a biography or an explicit memory about yourself in order to tackle this this aspect of this answer so so indeed massive States uh the there is alterations in sense of cell and it's up to debate what which um Dimension are affected by that and it's far from Clear what um actually happens to these kind of Imagine a more and many more other dimensional sense itself uh and we are here uh to unpack a little bit this we propose these quotations from Fashion 2008 uh where um it suggested that one in doing deep messy stage one experiences oneself and this very moment of Consciousness this is a quotation that may reveal some uh position about this um the debate what happened to the sense of self so in this case it's disargd that meditation reveal a very moment of conscience so there is something uh that is true consciences or uh something that is the real Consciousness and this actually has breaks to the concept of pure Consciousness so something that it stays there and uh um it's um purified by everything by almost to the body and there is so many uh definition of this pure Consciousness we brought some example like uh ownerless Consciousness the conscious that doesn't have an owner that doesn't have a self so it doesn't have as proper sense of self that walked or that owns this constant uh nothing it has worked on the minimal phenomenal experience defining it a special a temporal dimensional consciousness and this is uh very supportive because it's consistent with the Buddhist philosophy philosophical standard of no self so for the uh most tradition of Buddhism the um there is no self the self in is an illusion and it comes uh to our psychology when we enter in contact with the world and with the sensations of the world and certain uh patterns of behavior the analysis the tension we have a self so as it's that many patient it allows us to access this pure Consciousness for for supporting to this thesis and reveals one to choose an essential nature or subjective and we bring this into debate uh there are also uh people that argue against this and it's called Uh it has been called by Imagine their special issues the ubiquity thesis so the ability thesis claim that essential cells is ubiquitous to all states of Consciousness and sometimes it appeals to the immunity to error through misidentification argument of self-ass position reports which is simpler than this long name um and it's basically the fact that I can say that I had a wonderful

experience or I can and this wonderful experience may be true or false this is uh this on this weekend uh I can do mistakes I can make mistakes but I cannot not say that I didn't uh had this wonderful experience because the moment in which I report uh I have a perspective on this so I cannot really be mistaken on the fat end reporting this maybe I was dreaming maybe it wasn't a wonderful experience but it's me there remember party so the moment I'm reporting this it means that I have to be there and this is uh for uh applies also to uh when I ascribe myself a selfless State I had a self-estate this this is very controversial other problems on top of that are cross-cultural problems also addressed in the leadership so this experience selfless state may be influenced by uh certain vows or certain beliefs that meditators are endorsed uh especially in during especially expert meditators and the other uh thing is uh these uh all um cross-cultural analysis and debate about yourself maybe also uh actually due to our verbal misunderstanding so that the western western employee may have problems about the self that actually the Asian probably philosophy is not really addressing so there might be some incommensurability of of problems there but as also we said in the in the paper uh it's not there we want to address the Buddhist philosophical problems but rather we want to bring inside from from these two to our to our problems so the um given this debate uh we want to propose a shift in the focus so we don't want to start by looking at Consciousness and trying to solve things about the self but we want to start by the by focusing on the body so we proposed a shift in focus in the bodily experience examining this uh the imaginative state so let's start by the subjective experience of the body or the study or this ability to experience of the body which is phenomenology uh in the tradition our body has a unique stages so the anatomical structure of the body uh has been called by user cup so the living body the body that I can observe and others can be observous is the object among the other object they would say but it's not only that because uh it's an object that I can live from inside so it has been called the live the live body or the body as I leave it from inside as I said so what happens uh to these kind of Dimension again we can unpack this and see how the meditative States affect them so from the point of view of the life believed body uh we already know and I will unpack a little bit more so we have already said that uh we have called unlimited body experiences uh the three parts that spoke about this um fading of the body boundaries so the body bungee has been defined as a cognitive emotional perception of uh one's body surface that just finishing we can also debate about this but well it's important for us that our sense of body boundaries is flexible and dynamic it's not fixed because we are in the realm of the body as Fells from inside there is an interesting study that exemplifies this concept in relation to meditation in its uh by

John Brand in 2016 though it has worked uh late further on on this um as well but I find it interesting to look at this image because this is part of a self-report management that they use on a sample of people that has been trained in meditation in particular with the body scanning so that people had to systematically scan sensation rise in the body and the result is that in the less than four months of practice so no actually not the expert they already started to report an increased tendency to go towards the almost imperceptible perception of the body so they conclude that um meditation can be actually studied in this sense where it affects our sense of body boundaries and uh they say they saw significant uh results also in relation to increased mood happiness and decreased anxiety and regulation but this is uh yes another story that we'll come back later so I find interesting also this quotation by I think uh here in this paper is talking about why in good dissolution may be better understood as ego expansion and then um and then either the solution becomes an a later uh interpretation um and so I will read it it's it's not the feeling of self that dissolves with the boundary between a feeling of self and the feeling of anything else in so here again I think he's talking about consciousness but from our perspective this conversation is pretty uh informing because it goes in our interaction as well because he says it's not that you're losing really yourself is this this experience is uh dependent on how you actually perceive your boundary with uh you and what you don't consider you so self or fading of the self other distinction ability here it comes uh processing acting from framework because we want to basically breach these phenomenal subjects you experience to having this phenomena and underlying mechanics explanation uh indeed the processing activities I have recently used for this purpose also meditation research and uh Psychopathology research and so on so in this presentation I will use them in an interchangeable way but following certain uh Trends but it's not it doesn't have to be it's also under debate um and I will also give if I uh purely I don't know how to say um theoretical perspective on that I will not go too much technical um so briefly uh this framework considers uh brains as predictive machine or as a freezing likes to say as a considered brain is a fantastic organ uh which is because not it's not only uh able to do amazing things but also because it really generates fantasies about the world and this is because the in order to act interact with your with Excel and the world around the brain constantly uh anticipate the few anticipate the state current state of affairs in order to make sense of the sensorium or the stream of experiences that um so on uh so it possessed or embody or uh utilizes it depends on the commitment uh a generative model which is a model that keep tracks or regularity of the world and forms prediction out of it in a hierarchical way so highly reproduction uh goes

down to match uh sensory data um and sensory data goes up to uh um map to affect the predictions to update the model however every prediction comes with a degree of error which sometimes is called prediction errors so this predictionaries the discrepancy between prediction and um sensory input so that a new model is generated and uh they say the um first imperative or the only imperative of the system is actually minimizing the prediction errors so that the model can most efficiently uh cope when interacting uh and act in the world however it's not um the only imperative because the as I will introduce in this Slide the the system uh has also to estimate the confidence or likelihood or Precision of prediction in order to evaluate the prediction errors uh that are coming from the discrepancy so that the generative model has to compute enough a fine way uh how how much Precision to assign the to the prediction or prediction errors and I choose this image because it renders well how the distribution are um set depending on the prestigious sedation so if you have a precise medication error then the posterior distribution will be closer to the prediction error well the imprecise prediction have lethal effect on the prior so that the procedure the the pursuit distribution will be closer to the prior and this is very important because if the um that Precision estimate are not weighted correctly then the system generates false inference that will not able to the system to interact efficiently and update model efficiently so so far we have uh mostly talked about uh prediction error minimization uh by updating uh the model however the system can also uh active samples or active modifying the prediction error of the world um in order to fit uh the prediction errors to the world and this comes into play active inference um that builds upon the free energy principle where uh that um is based on the idea that living organism maintaining themselves with an optimal State set of states that comes from the second low dynamics of shortening Schrodinger so given that it comes into play uh how attention is analyzer examine it under this framework so given that uh the organism has to maintain uh Within These optimal assessor states by maintaining an optimal balance between uh computation uh and precision optimization SO waiting between uh prediction and um prediction errors attention comes into place as a modulation of these Precision of sensory data working as a Searchlight so in other words it optimize the signal to noise ratio it's the one of the way that we have to uh put in place in place these local optimization of precision estimate and this is very important uh why we bring into attention because meditation has a systematic energy reservative use of attention resources and we will come back into that further on in this presentation uh and given that we are interested in okay we have seen the live body what happens to the funeral experience now we're interested in what happens to the corporate what happens to the living body that can be

observed by the the organism that during medicine States in these bring we brings into play in other two uh important concept uh in active immigrants which is a way to uh self-regulate the underlying all and what experience Soul subjective experience so two ways uh or one way that then is also active so the first way is homos little homostasis which is the maintenance of required optimal Say by survival and reproduction of self-organizing biological system so for example I'm cold and I use my internal resources to keep my state into this into optimal state so I start shivering however I can also put in place uh active modification in the world so that um I'm not cold anymore so for example I take a blanket this is called elasticity so it's active engagement and modification about the environment in order to maintain myself in optimal States for survival so we can say uh of course human bodies are self-organizing biological system and aiming at keeping these internal States within a certain uh viable range for survival now given that we will come back again to phenomenology off touch I will go I'm going to do this for the second time uh to then wrap up everything again uh so we have talked about film all video of the body however uh we haven't uh underlined important is that the touch and proximal sense has for our lip body our as Technica and colleagues has already proposed our proximal sense are Central for our minimal sense of self and on this on on top of that we bring important to play the fact that different kinds of touch can convey different experience of this and so stuff or sense of boundaries so for example imagine yourself that you are um in a car of partial or California and you touching the whole environment with your body you may feel that you have a sense of immersiveness or that your body boundaries are slowly a bit becoming less clear and this experience it's actually a bit similar to the experiences of sensory deprivation States where people are put in a salty water and they let them flow for some moments sensory deprivation states are not really really all deprived because you have other activities in the body and movements that produce signals that you are actually aware of however in both these experiences the body boundaries seems to fade a little bit however if you're in the cnu's meeting and then you touch the rock with your hand uh you clearly will feel yourself as separated as uh something different from what you're actually touching like sense of separation things phenomenology can be both interactive inference through the mechanism of some of the sensory attenuation which comes from the sensory attenuation so this process happens when uh the system depicts your its own action uh so for example I take a glass uh and in this moment my Rena has already created a different copies of this action and the sensory consequences of this so whenever I grab the um these this class the sensory concept that happens to my uh but they are slightly um attenuated and this is why we cannot tickle

ourselves because the prediction errors comes at not surprising this has a implication for our embodied experience in the world because we are feel that we are immediate immediate content with ourselves in the world and this is because our country experience are mediated by transcript or non-perceived cognitive processes so that the system leads can leave behind or this attend self-related sensory processing so and these allowed to allocate your resources to perception and action involved in the processing of the object of Interest so for example when uh limousine as this example of the Cherry when I have to uh take a object that Italian for me I'll be much more interested in the uh picking the object and I will not really focus on my arm boring in picking that so that the system needs to compute and maintain a fine balance between external and internal comp components of some other sensory attenuation and this is very important for the sense of self as well so it uh we need to understand and result that we need but in order for us to have a sense of self we normally have a sense of what we are when we don't and what we are not but for the brain it has to uh or as as uh agents end up with the brain we compute this distinction so where the self ends and where do work begins or do I move my arm or my arm is moved by something or someone play the role in this uh distinctions and involved in building multi-sensory bodily self-representations so that the sense of the self and body is uh subjectively and phenomenally transferred that's that that means that when we act in the world and we interact we normally these attend uh the way we are in the world because we need to be in contact Direct contact with the world if we always attend it to our body and our our um ourselves in to in the sense that we will be hyper reflexive then we will be actually uh will be impaired in what we normally do and this is actually a condition that we will talk about later so that in our students in the sense of selfies and ongoing inferential process center in the body so what we propose is uh what is lost during selfish selfless experience is an ordinary way to mental mentally model self in relation to the body in the world so that if we think about the self as as I propose here then I uh the iceberg where the narrative cells is what we see but the renewal self is what we actually are composed of and it's actually larger but we are not aware of similarly when we have software experience we may have also these uh inferential process where we think we'll have lost ourselves um however uh we have gathered some evidence in the fact that meditation facilitate all genomic regulation involved in homostatic processes that means for example that meditation increase the synchronization of cardiac and respiratory system uh this is connected with reduced levels of blue blood glucose or blood pressure and heart rate variability so there is a successful medication helps to successfully self-regulated the body uh especially in comparison to stress

that actually does the opposite and on top of that uh meditation help us to attune us to our physiological interceptive awareness uh and this um it does it so they say by enhancing the processing of extra effective interceptive stimuli which it doesn't have to be understood as a maladaptive uh increased body awareness but rather a adaptive body awareness which is very distinguish distinction that we can come back later if you if people are interested or they ask about these evidence are also in line with evidence communication that promotes self-regulation of pain and pain and pleasing and so three-day attentional training the in allow uh you're actually able to self-regulate the pain and not not really like the sensation of the pain because it actually gets very clear but the uh automatic behavioral patterns that comes uh in association with the with the pain so there is also also self uh regulation on emotions and so on so the body doesn't know literally disappears actually it stays as a very centered uh as a first year at the center as we propose it in the paper and I proposition through this image at the center of a cycling so self-regular self-regulatory processes allow the subjective experience to lose oneself so the percent of the body remains present in the background and these for example can it's very here from these reports from a person undergoing this kind of experience that they uh say that there remains a sense that there is a body without any experience of a sense of boundary this is not very last slide but we come to our wrap up of our claim uh attenuating the self unlimiting the body which is also the title for our paper so starting from the fact that all medicine practices across the three Traditions require systematic and parallel fully crafted body preparation to enter these deep method self-state self-estate so it doesn't happen that all of like all of a sudden you just have a selfless state which actually may happen more similarly with a psychedelic bubble meditation you need you need to pass through gradually many different uh state or many of different level of concentration that happens when you have around a predictable environment and you do so by building prior expectations so you have to keep your body in a certain position of posture as you say this in this image the person is seated with a very straight but not very but a straight back relaxed this is very important realization when it lags and so on and then you need at least uh in in the most part of your practice you need safe predictable environments with temperature that are enables you to maintain your homeostatic analysis and processes so all these conditions are necessary for your successful homeostatic self-regulation in this sense when this part first part of the preparation to Andrew then after the state of concentration the meditator can let go and leave behind self-related processes and process the body transported or it can attenuate them just attenuate its body the body cannot can be attenuated so our claim is that a transfer essential cells and body

May underline the phenomenal phenomenology of losing one's self and blurring body boundaries as reported during meditative so our conclusion is that we have proposed a shifting focus on homeostatic body self-regulation in examining the majority States the experience of having ability a living self-organism biological system is never lost in this process rather it stays transported at its very center self-attenuated yeah very much present in a handful life so that the self model is attenuated to a big week to become transparent enhanced processed in the background now giving this conclusion now I'm gonna present related project the first one is still uh on the theoretical side so we are in the real of active inference model in meditation research there is a radio literature about that um that um base that have studied different kinds of meditation techniques like focused attention open monitoring non-dual awareness um they have for example uh showed that these uh attention techniques profoundly and alter typical mental experiences for example Focus meditation is the um that engage attention on one specific object experience for example the breath uh and in active events there we mean that Precision is uh estimated one century input which means that it reduced the impact of the other thoughts and parallel processes that going on or often open monitoring meditation is the uh is the engaging of attention to one's experience as a whole so the whole stream of experience with us selecting anything in particular so that in an active intertherens it has been described as assigning equal Precision to all these experiencing promoting non-reactive sensing and it is connected also why meditators are good in self-regulated regulate pain so in we will also build our model our us on self modeling based uh on the previous literature which is in preparation and then going experimental uh we want to bring into the table uh applications of what we we're researching and potential benefits may be for a case or people or yeah individuals that suffer from disrupted body awareness uh or disruptive self-awareness and there is a relatively recent condition that has been studied that is across the personalization so the personalization um is their destruction of both body and self-awareness and it's a very heterogeneous condition there is a lot of different dimensions uh that occurs but generally uh it's connected with feelings of Happiness disconnection feeling the body isn't the as an object rather than a transparent medium of experience uh which is uh switching release called disembodiment it is linked with difficulties in concentration and attention often comorbid with depression rumination and anxiety and recently and colleagues that also propose a uh characterized by failures in attenuating so much a sensory signals based on what we have uh previously said and based on how these um condition affects uh a large number of population we want to test certain hypotheses so the first one would be the highest somatocentric attenuation correlate

with more feeling of aliveness and wide openness in experienced meditators and rather lowest from other Century attenuation could relate with feelings of unrealness Badness in people experience uh decorationalization and on the other hand severe homeostatic dysregulation or bodily States during deep Methodist States may lead to negative emotional outcomes and other self-experience this comes from to the fact that the personalization and meditator and meditation has some has uh can both create stressing negative outcomes and it's interesting to see how these happen and what are the commonalities and differences in that and also we have now in the realm of spatial temporal perception which is uh connected also with mood disorders or like depression and anxiety uh and also with the personalization uh however in the precise station there is no specific items in the questionnaire that is usually uh used for measuring so we're which is a Camry Cambridge the personalization scale so we want to test this as well uh and we want to test it uh in people reporting selfless and disembodied experience so both pathological non-pathological conditions so path of this hypothesis we are already going through this is our team coming from Humboldt University where I come from the month but also a university Hall and College London with anus and Joseph first study is as I said about time perspective so we want to analyze uh um how people with high rates of the crystallization can relate with dimension of the time perspective so there are more future oriented or more negative oriented our hypothesis that they're more negative past oriented so they're stuck in a very long present moment that but also the the their the tendency of ruminating so always going to pass negative memories this is a title lost in dark times the effect of the personalization experiencing on time perspective and these we are already in the analysis of the data but it's on ongoing still study so we we have preliminary data on this uh while studies two we're in the recruitment phases but we are almost there and encompassed the uh hydrates or people individual with high traits of the personalization and meditators I want to see through a set of questions and a behavioral task how they may differ in the sense of self body and motion of course they're very wide and conflict Concepts so there is always room for for feedback and replication and so on uh we want to just we wanted to we have selected this tasks the embody task that I've adopted from newmanma uh where individuals have participants has two colors parts of the body in relation to activation that they feel by reading certain emotional story so these stories at least are supposed and validated to at least see certain emotions in the body and then people colors parts of the body and then we will see how people with high traits and visualization limit heaters will will different this and we will also compare with other questions that we have put uh potential title is emotional body maps

of solid Detachment personalization and meditation and with this I thank you all and that's it now we are open for discussion thank you awesome thank you very much Valeria your music down thank you Valeria great of course well very fascinating presentation perhaps uh on up first and then our other colleagues who have joined would be awesome to hear your after be here on this work and and any reflection or questions you have on it is it my turn sure did I missed yeah yeah so um obviously I don't have many questions because it's like we brought this together so but um I I think what I'd like just to add that really a question is I really like to see um so for instance like in um an ideal world would be nice to you know uh get all the the tradition of the meditative experiences and how they relate to the body and make a very precise prediction on what exactly they will kind of like predict in terms of uh certain uh sensory experiences and how this can be interpreted through different like theoretical lenses I think that could be really really interesting because I think one of the problems uh the challenges that we have right now is that we have so many Concepts so many approaches and kind of like scattered all over the map and I think would be really nice to um you know a narrow down because as you said Daniel um so everybody knows what meditation is everybody knows at least one person meditates in in life and everything but uh it's really unclear what exactly we are um we are after and why in certain cases this is kind of like um beneficial and one in another cases basically it has a negative consequences and this is why precisely we are doing this our project on the personalization and meditation and I can share with you informally scoop but data is here so it kind of like I can share with you so what we did in this project basically we took people with typical controls that don't have the personalization and never meditate it right so it's kind of like yeah never did that and then we took people that meditated but um and then we looked at whether they have kind of like high or low traits of depressurization and then we took people with deeper sizes that never meditate right um so so I have this like um inclusion in exclusion criteria and um yeah it's it's really interesting to find what our finance is like it's really difficult to find people that have experienced meditators and low depersonalization skills low depersonalization traits right so most of them they actually have ID personalization but then it's interesting to see how that contrasts with like the people that have the personalization and no meditation experience right um so that's that's something I would like to um Carry On and um yeah so to reach the community and um yeah and that's something we will definitely carry on with Valerian the next steps and also with some people that I have here in the chat uh Julia and Simon so Simone will going to work with us on also smarter sense retaliation yeah through different

senses and Julia we have some um without scooping we have also some very very interesting results coming up um we just finished data collection on intentional binding and depersonalization so related to the sense of agency yeah so yeah so that's all from here just like for now we're kind of like in between because well I represented very well like the theory part and the studies are literally just getting hot from the oven now and we hope that in September it's like we're going to get the mode there yeah but the data is in the box so that's very exciting and yeah we have the results yeah thank you Danielle again for uh this platform it's really nice and as I said I really hope that after this discussion we'll have more um um you know opportunities to change exchange and um yeah discuss we also plan to organize something around a duration of self-consciousness in across various conditions and also the personalization in Elizabeth next year so maybe you can cross the pond and check for yourself that we have a nice airport in Lisbon cool well good to uh get the appetizer before the entree so that's very awesome about just how much Theory and um even unconventional scientific theory like World Knowledge traditions come into play and then what it looks like to take that experimental step to make it scientific is is definitely what I saw on the talk um Julia or Simone do you want to give any reflection or any thoughts or questions yes I actually have a can you hear me yep yeah I actually have uh three questions I know it down um I think the first one because it's just for me personally I'm curious about that is that those deep meditative selfless States I think that can also be reached while doing yoga right basically the opposite of like sitting still is like a sort of repetitive movement actually so I'm wondering do those movements have to be perfectly predictable in order to be attenuated and then reaching the same state or is that like a different uh different concept or how do you how do you put that in the in your in your framework mainly to uh to talk to topics one is is it possible to reach deep Methodist States also uh not sitting in a steel position and the answer is yes however there is not so much research on that so I can talk about anatical knowledge but I did not read so much about uh World communication effects usually works meditation or yeah of course you talk about yoga I know but that's what I uh would say that walking meditation is used uh as a complementary practice through the same thing so I would say hard to reach deep Melody stage only by World communication in regarding to yoga also yoga means Unity so what we think about yoga now we think about movement but it's always uh occupied by reduced state of conscious concentration through meditation so you cannot have yoga without meditation otherwise we'll be stretching your glasses but uh second topic is that can it happen when you have when you don't have prediction of movement right no I meant like because in yoga I

mean at least I don't really do yoga right but I can imagine that what when people reach those States doing yoga then they kind of do it almost like in a in a trance type of way they're always doing the same movement over and over and over again so that they don't have to attend to those movements but it's just like an automatic process I was wondering if it if that also can I can I jump in so um um actually this is precisely the opposite what is what what Valeria was saying is so but I was saying this is precisely the opposite you want to reach through yoga just like the opposite of automatic Yeah so basically when you do this movements you need to be United with your self-training it so you really need to um yeah uh so yeah I'm doing yoga so they they the past 10 years and precisely this is like they're trying to tell us it's like don't don't do it on automatic because it's like when you have your thinking happening there and your body is doing some stuff there right you need to be basically be presents in every single type of moon so you have like a fully fully a presence if you do it automatically that's like that's more like gymnastic or something yeah so um but it's interesting uh so your point is really interesting because then would be really nice to have a control condition where we have people doing like movement meditation and yoga and then um contrast with the phenomenon of like um uh some other sense retineration and uh whether they have what type of light embodiment and disembodiment adjectivity experiences they have okay but yeah I don't know exactly what I was trying to get it but I think it was also more related to because you said that when meditating you have to sort of let yourself letting go exactly the ideas then everything has to be perfectly sort of predictable right that's that's the idea behind it so then I I'm I'm thinking so you always within yoga don't you have to do like the exact same movements in order for it to be predictable in order for for you to let go and not over attend to the environment does that make sense or if I thought when when were you both uh on the Simo we're speaking I think on one side you see more maybe you're referring to the implicit procedural map implicit procedural knowledge that can uh you can acquire through yoga and you and I are talking about doing things without being aware which is also automatic and left and you both are right I think in this because on one hand if you and you would say like if you do yoga you learn how to to do yoga right so you learn how to uh move yourself in this in this way so that when you do a certain position you know then you have to arrange your body and for a ranger that you always have to keep monitoring yourself so there is a continuous going back to your body keeping track of your body and continue to monitor yourself but in this way because you have learned to do this these um constant constant going back and being aware on yourself and these comes also when you you do your movement you have learned that you

certain posture are made in a certain way so that it becomes more or less predictable actually what you're gonna do right but it doesn't mean that you you don't have to be aware and actually there is a trick uh reload here because you're a very attentive with what's going on but at the same time the more your attentive the more so much a sensory attenuation may happen and actually it's what we are trying to argue in our active uh inference model the the more a predictable it gets the more it gets attenuated but doesn't mean that you're not aware that's that's uh yeah that's good thank you for this input because it helps us elaborate this I think it's super interesting because it's like when you when when you look at like people that climb or surf or whatever any type of sports right in the beginning in order to learn the skill you have to put a lot of attention to the individual movements but then later on in order to actually be good at it you have to take the attention away and then in those moments is when the people report that they're actually feeling sort of selfless days in this flow States right which is when the the movements are internalized so to speak I think that's that's super interesting can I can I just shoot the other two questions I had they're like sort of different or should we go to Julia first maybe it's true yes I don't know I sorry um yeah sorry for the lighting and sorry because I'm in the lab so it's probably not the best position uh thank you very it was very interesting and I'm really uh looking forward to um yeah look at the uh yeah the results you you have in your study especially yeah I mean since uh as and I mentioned we had something interesting with the international Banking and the personalization and um yeah kind of happy about that and what I do so I'm really sorry because I listened kind of doing something else unfortunately in the lab I was busy um while you were talking but I kind of like the idea of yeah that's the self is attenuated or not or increasing and kind of goes back to just Simon the last comment from Simon about expertise I think in terms of motto uh interaction there is some literal tribality expert I was trying to Google that um that I attended a workshop last year and I think it was about once you're an expert you at the beginning you have this kind of flexical you you you have a better understanding Consciousness and then at the end you kind of learn to let it go to write the flow of the motion kind of so it's not that you are and I think it's probably maybe the same I don't know whether yeah but attention in meditation is that you have this maybe more like not focused uh it does not stay but more like open um sorry I forgot the name of the meditation type of mission you know but open meditation and monitoring yeah yeah exactly so you I mean I don't know it's uh so it would be interesting and I suggested to see what kind of meditation what type of meditation maybe change the sense of the self maybe at the beginning it's more acute because you really scrutinize your sensation and and then after I

don't know this is something that you plan to do right or should I go or do you want to go on uh the the first question was I know ah there is a body of evidence they actually say that in expert meditators uh attention to start decreasing but there is also because you allocate less and less attention but I would be careful with that because with with this because they say there is an uh decrease that uh decrease activity in anterior singlet cortex so they make inference about it um as as a meditators as well uh and having to know other people there are more experienced than me they would never say we lose our attention to that because our experience is so rich our Sanctuary signals and related thought process that you cannot afford to not be attentive however we can drink into the play different kinds of attention as you said there is focused attention or there's an attention that requires more explicit uh thought about that some more explicit kind of uh way to focus on things and there is more like a bear bear attention so when you focus on the whole experience and you let things go with a non-judgmental attitude and more open attitude and is actually this attitude we're very stressed in meditation techniques that's why they're a bit under lock and in research I feel because it's it's a lot about how we use attention but not the intent attention but not how we use intention so our attitude has the attitude that mediters has to be very open and very non-judgmental and these reduce the non-reactive incentive for for example and reduce the ability to self-regulate the your experience your life yeah I have partially answered you know I mean I kind of like the idea that you you linked that with attention and um it's just I think maybe it will be interesting to look at different type of attention because whether you know you have like so maybe there is a difference between I don't know different type of meditation May develop different type of I don't know if maybe the open um meditation he's more prone I don't know because you mentioned the Instagram cortex into me it's more associated with a certain type of attention like more like maybe um more prone for like a feedback um or negativity or these kind of you know error monitoring process so it's not like your attention you lose your attention it's just it's more like in the background and then when there is a mistake or something it kind of pops up more quickly I don't know and you may have more like uh frontal or a sustain attention I don't know I mean I'm not especially of attention at all so please take that little bit of self um yeah I was wondering if this is something I don't know obviously I mean because it's only you're not trying to do any EG or or narrow physiological measurement right yeah but in the in the pipeline yeah okay but we really want uh with um with Valeria is and other collaborators is to basically push against this idea that uh you can have selfless experiences I know it's controversial and many people talk about oh selfless

experience below uh so basically we want to say radically speaking okay though so that works if you buy the bullet in your mind what you do a list and somehow consider that actually the body kind of like disappears in the process but if you're not mind body dualist then you need to make basically put the body into the equation and realize that actually what allows the mind to disappear is the fact into the pureness of the experiences the fact that the impure body is kind of like put in a stable condition in such a way that basically can afford to live it in the background and yeah to have like this type of like um experience arising but without that in place uh you it can't cannot really afford to you know go through the selfless experiences and this is why we want to be provocative in the titles that well actually maybe the experience of like selfless um you know selfless experiences actually comes necessarily as to sign of the same coin of this experience of like having a unlimited body that actually expands to through the world but the body is there I mean the the thing the The Entity that has the experience of no experience is that a body that kind of like is um self-regulated in a certain way like the homeostasis is put at like a set point and when you're using this metaphor of like think of um a cyclone Right Moving Cyclone so you have the steel eye of the cyclone in the middle doesn't mean that actually is not there it's just like it's around everything and because we are self-organized in biological systems uh we kind of like this animal this thing needs to keep track of this internal bodily states and their I mean just a successful system does this all the time all the time literally from second one when we basically get from one cell to you know the substanceism some sort of like fascinating self-organization that is happening on the cellular level already and it like needs to be yeah and death is basically this is what is happening it's like a single disintegration of like this um you know self-organization and um but that's clearly that's not what is happening on meditation right because you do hopefully you don't die when you meditate in that sense there is a disintegration of a certain way to experience yourself in relation to the body in the world but clearly there's something that stays there needs to stay there for you to have those experiences so that's a bit of controversial radical claim that we like to defend and uh yeah we already had this discussion several people for instance like whatever and I we've been to uh Enlighten and this workshop on um of knowledge and neural computation we met there with people like Thomas betsiger and everything and they always have a different opinion so kind of like have a lively discussion about why we think it's like what does it mean to actually Lose Yourself and what is the self and what is the pure experience and why we basically need whether we're like it or not the connection with the impure body to make sure that it can have at a certain point of pure experience but then but

then again this is something that just speculation at the theory level and this is why I have this brilliant team here that we we can do some uh studies and actually measure stuff put the finger on it put the number yeah and see what we get these are these are awesome points and there's so many ways it could go I'll read a question from the live chat so G pagnani wrote the notion of increased somata sensory transparency in experienced meditators associated with increased liveliness of the experience is intriguing can you say something about which levels in the hierarchical structure of the internal model are more likely to be affected by the meditative attentional tasks in terms of precision deployment do you think the attenuation of somatosensory prediction error concerns more intermediate levels in a way increased meta awareness rather than sheer sensory attention I just wanted to ask rather than ensuring but rather than sheer sensory attention yeah you can go on now please yeah so uh yeah so this is something I I'm I'm trying to develop this idea right now with um a PhD student of mine um uh Libby uh Liberty service basically uh and this is why it's connected with the question that you ask you'll see is that um you know your platform is active inference right and so people are kind of like you know um lately kind of like realize wait a minute actually we're active and we need to explain action in the world and everything is kind of like comes late right but uh if you look uh going back to the animal basically um the action and the movement is there from the very beginning so this means that this is why it's so important to basically relate the um um you know self-organization self-awareness self-perception necessarily related with movement not so movement so on action is not something that you add on the top of this right like uh and some people like intention so you can have an intention but basically the first movements are kind of like automatic and need to be there right and basically our body moves all the time because we breathe all the time and our heart beats all the time so this means that for me this type of like attenuation it's happening at the sensory level because that's the basic one so can you have sensory stuff without metacognitive stuff yes can you have the cognitive stuff without sensory stuff no so it's like what what is basic that's the first things needs to be um um uh alternated and then then it has to be a balance between meta cognitive and sensory attenuation because uh this is precisely what I think it is happening in the personalization here because if you if you focus too much right so if you lose the the balance right between then you you basically yeah it's it's kind of like this type of you cannot afford to sell forget yourself yeah and then it's kind of like the self-standing way and that gets to get you even more disconnected so to speak yeah uh so this is why I think it's important to um uh contrast different type of like a meditative experiences and meditative practices to see which one of

them actually um has more focus on what you do with the metacognitive metal Oneness awareness of awareness and which one are actually involving more like sensory movement for instance like let's say yeah so that's that's something that needs to be um empirically tested but I would speculate and then obviously we can put a test either by doing a simulation or you know actually collecting data to see how that um uh plays into this uh distinction I think Simon wants to say something but I'm not the moderator So Daniel you decide yeah Valeria first and then um okay uh yeah thank you Anna that was very uh I don't have so much more to say because I agree with her but uh I thought when Jani and you were reading the question I for me the answer was more or less clear in the sense that um as I say the somatician continuation happens are the very low level predictions so in a very uh pre-reflective self-consciousness uh so that the person was asking uh if you know at least in the meditation techniques with the when you put the attention on for example your bodily signals there you have a top down modulation of this uh of precision racing but the top 10 modulation is on low level predictions so you're not putting top down on what you think about your body is you're actually doing the opposite you're trying to forget about your uh previous expectations about the body because you're trying to have this non-judgmental awareness about what's going on in your experience and then uh with this um let's say uh less uh biased uh way to attend your body you start piercing and perceiving uh bodily uh Sensations in your body or yourself or other kind of experiences so I would say this increasing Precision on somatosensory signals happens at the low level of the hierarchy um that then attenuates the sensory consequences of this um of the first year and the math awareness comes uh when you learn the when you're I think when you start having increased body awareness so you you'll make more experience I think then you start developing meta awareness about how your body is actually about higher representation of the body of your body but in the moment when this happens uh I think it really happens that the low levels of personal beliefs and then later on the longer time scales then it starts becoming uh more abstract and ingrained in your personal uh at the personal level of your beliefs that's how I stay but of course uh if Simon has something to do with it um yeah it's really open I know I didn't I didn't want to comment that I just selfishly erased my hand to ask another question Yeah okay or Daniel I don't know if you have an opinion on that I'm happy to hear it as well well certainly resonated with uh this comparative approach to studying the the diversity of methods and protocols we already have for mind and body and attention broadening the scope and just kind of pointing putting darts in the space of of what practices do people engage their mind and body in and then how can that territory be explored with also

a variety of theoretical models and how does that relationship connect to the kinds of empirical work that we might do in a brain scanner or um with other kinds of measuring um and I think one thing I I I want to explore more is like in a hierarchical nested cognitive model which we often speak of ourselves and other uh types of sophisticated entities as where's our awareness where's awareness Matt picture and um and and also to Simone's question about um when we're moving but we're expecting what we're moving then how does the exteroceptive flow like the visual field you might be very familiar with walking and so you might be able to proprioceptively kind of zero out on your walking but there's this extra receptive um flow of novelty in the world so I think that that how um and then connected back to the comparative and staring at a candle or a blank screen or darkness and all these different methods it's it's there's a lot that that these questions bring up thank you Daniel so I I have something to add to that because I think it's super important so um so for me it's not so you know my background is like in body cognition and activists and everything and for me it's really not at all surprising the fact that many of these techniques have been developed by people that kind of like lived in closed communities right so it's like you have a space where people can you know meditate walking I don't know and then they have this ritual they wake up at this time and they do this thing so you put yourself in a yeah it's a ritual right like all the to reach also so this means it's like as I said it's like high predictability um and then that's that's something super super uh important yeah and even if even in the walking meditation you need to um follow a certain path um uh yeah so that's that's super interesting well um in in the last uh five or ten minutes I would love to hear from each of you like what are your directions heading forward and and what's the um what's exciting to you and where would you like to take this maybe um Simone and Julia first and then we'll kind of make our way back to the authorship I mean I mean I'm not not myself not working on the project right but like what I what I was really interested in sort of extending this is trying to understand um because the way the way those deep meditative states are reported sometimes and the way um people with depersonalization disorder report their experience is very similar they almost did like the same words right and to describe how they feel and I wonder why one is experienced positively and one is experience so negatively as in it being a disorder and completely disrupting someone's life so that was like this this comparison I think I would be very uh curious about awesome Julia if you want to add anything um we're not it's it's very interesting and um yeah I mean again I'm not so involved in this study so it's a bit hard but I mean I'm really interested in the I mean all the Selfies affected by yeah our interaction with the world external world so uh and I'm

completely already spoiled with the result we have with the international binding so um uh it's very interesting I mean yeah it's the self is attenuated or um if it's disappear kind of I think that's the key uh question we want to ensure and uh yeah I would be very curious to know I mean what very I found in uh data uh to see if we can have a common narrative about what's happening um also with the personalization um yeah it's very interesting to see how moving or body I mean attention uh to the body can really change the self kind of for the way we process sales information um yeah so very interesting or honor yes so um there's like lots of work to do ahead but that's exciting right yeah so very short term it's like we need to write up this results that we just got um so yeah in relation to the body so basically going to have some numbers to show so that's that's super nice um and then the next steps would be to um you know create or develop a computational model around this yeah with basically contrast you know the real world data with you know um simulation data and then uh yeah so have a multi-layer type of like approach with different methods from physiological to behavioral to phenomenological to computational potential also neural so that's something with would be nice to do in the next steps yeah um so yeah um and apart from that I think that's all for now uh and I'll let uh I'll leave uh the flow to Valeria well again thank you again and I know everybody uh yeah I I also wanted to thank you a lot for this platform this invitation it was yeah really nice a really nice person and really nice conversation I'm working with on in these projects so she already said what what's the the next Direction on what I can add in this I think would be on the theoretical side also start engaging with more closely to bodily reactions to meditation because the body boundaries the solution is just one part of it it's like one part of the phenomenology but it's not the only thing that happens during meditation um there is a lot of potential that can be discovered uh on this because because the body boundary is the solution again it's really study a lot but it's a phenomenon and there is so many other things that happens in our body while we meditate and you I I speak for first first person experience but in the community it's really like already sad other kinds of also um uh more healing Traditions like breath work for example there there is a lot of potential where people already work on that but there is no Bridge uh with the academic communities so people work pragmatically with this intuitive knowledge as well what happens in the body during these alternate States Of Consciousness but still we need to rage it with uh trying to make sense from a phonological and scientific point of view an experimental point of view as well so I think I see myself in yeah trying to step in this tour then like bring Insight from one world to the other and opening uh to help myself and the other people around me yeah awesome well

it's the Vlog with the work that sounds really exciting love the idea of the the two-way street with different bodily and cognitive practices and collaboration and conversation that can bring in all these different expertises that that maybe don't have an fmri study whatever that's worth thank you though always um any of you are welcome back and really appreciate that you all came out thank you Daniel thank you see you soon